Dengue notifications in Brazil and Rio de Janeiro

A presentation-ready DATASUS/SINAN workflow

Purpose. Describe the official weekly dengue-notification patterns in Brazil and Rio de Janeiro, then show how same-period Google Trends and Portuguese Wikipedia attention signals were evaluated as retrospective 2025 nowcasts.

Structure. Part I stays with the downloaded DATASUS/SINAN outcome series. Part II adds the archived online-attention signals and model outputs already produced in this workspace. Notification records are the outcome; Trends and Wikipedia are predictors.

Presentation flow

1. State the surveillance question and define the outcome.
2. Load the four archived DATASUS/SINAN weekly series.
3. Check dates, duplicates, and the incomplete boundary weeks.
4. Compare Brazil and Rio in 2024, when the large epidemic wave is visible.
5. Compare the 2024 and 2025 seasonal patterns within each geography.
6. Close with descriptive takeaways and limits of interpretation.

> **Key definition:** each value is the number of dengue notification records with symptom onset in a Sunday–Saturday week. These are notified records, not a final confirmed-case count.

In [1]:
```
from pathlib import Path

import matplotlib.dates as mdates
import matplotlib.pyplot as plt
import pandas as pd

ROOT = Path.cwd()
RIO_DIR = ROOT / 'rj_trends'
BRAZIL_DIR = ROOT / 'wikipedia'

plt.style.use('seaborn-v0_8-whitegrid')
COLORS = {'Brazil': '#2563a8', 'Rio de Janeiro': '#b22222', '2024': '#b22222
```

Step 1 — Load the official case series

The national files are saved in `wikipedia/` because they were prepared alongside a separate pageview exercise; here we use only their `date` and `sinan_notified_cases` columns. The Rio files contain records for Rio de Janeiro residents.

In [2]:
```
def load_weekly(path, geography, year):
    data = pd.read_csv(path, parse_dates=['date'])[['date', 'sinan_notified_
    data['geography'] = geography
    data['year'] = year
    return data

brazil_2024 = load_weekly(BRAZIL_DIR / 'brazil_sinan_dengue_2024_weekly.csv'
brazil_2025 = load_weekly(BRAZIL_DIR / 'brazil_sinan_dengue_2025_weekly.csv'
rio_2024 = load_weekly(RIO_DIR / 'rio_dengue_trends_vs_sinan_2024_weekly.csv
rio_2025 = load_weekly(RIO_DIR / 'rio_dengue_sinan_2025_weekly.csv', 'Rio de

weekly = pd.concat([brazil_2024, brazil_2025, rio_2024, rio_2025], ignore_ir
weekly.head()
```

Out[2]:

	date	sinan_notified_cases	geography	year
0	2023-12-31	45848	Brazil	2024
1	2024-01-07	61515	Brazil	2024
2	2024-01-14	89814	Brazil	2024
3	2024-01-21	127227	Brazil	2024
4	2024-01-28	164645	Brazil	2024

Step 2 — Validate what will be presented

Before plotting, confirm that there is one weekly value per geography/year, that case counts are non-negative, and identify the endpoint caveat. The final week beginning **2025-12-28** crosses into 2026, so it is omitted from year-to-year comparisons. The 2024 file also includes a zero-valued week beginning 2024-12-29; the overlapping 2025 extract provides the usable count for that week.

In [3]:
```
quality_check = (weekly.groupby(['geography', 'year'])
                 .agg(weeks=('date', 'size'),
                      first_week=('date', 'min'),
                      last_week=('date', 'max'),
                      duplicate_weeks=('date', lambda x: x.duplicated().sum(
                      negative_counts=('sinan_notified_cases', lambda x: (x
                 .reset_index())
display(quality_check)

assert quality_check['duplicate_weeks'].eq(0).all()
assert quality_check['negative_counts'].eq(0).all()
analysis = weekly.loc[weekly['date'].ne(pd.Timestamp('2025-12-28'))].copy()
```

```
analysis = analysis.loc[~((analysis['year'].eq(2024)) & (analysis['date'].ec
print(f'Presentation analysis set: {len(analysis):,} weekly geography-year c
```

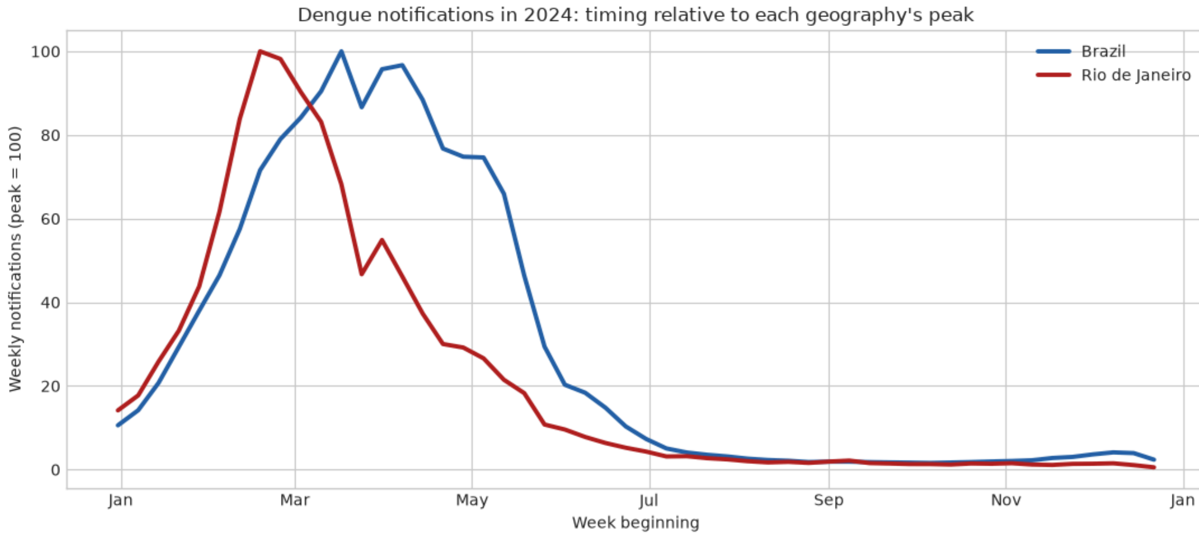
	geography	year	weeks	first_week	last_week	duplicate_weeks	negative_c
0	Brazil	2024	53	2023-12-31	2024-12-29	0	
1	Brazil	2025	53	2024-12-29	2025-12-28	0	
2	Rio de Janeiro	2024	53	2023-12-31	2024-12-29	0	
3	Rio de Janeiro	2025	53	2024-12-29	2025-12-28	0	

```
Presentation analysis set: 208 weekly geography-year observations.
```

Step 3 — Establish the 2024 epidemic context

Start with 2024 because it makes the shared seasonal shape most visible. Brazil and Rio have very different population sizes, so this is a **timing comparison**, not a comparison of burden per resident. Each line is scaled to its own 2024 peak (=100).

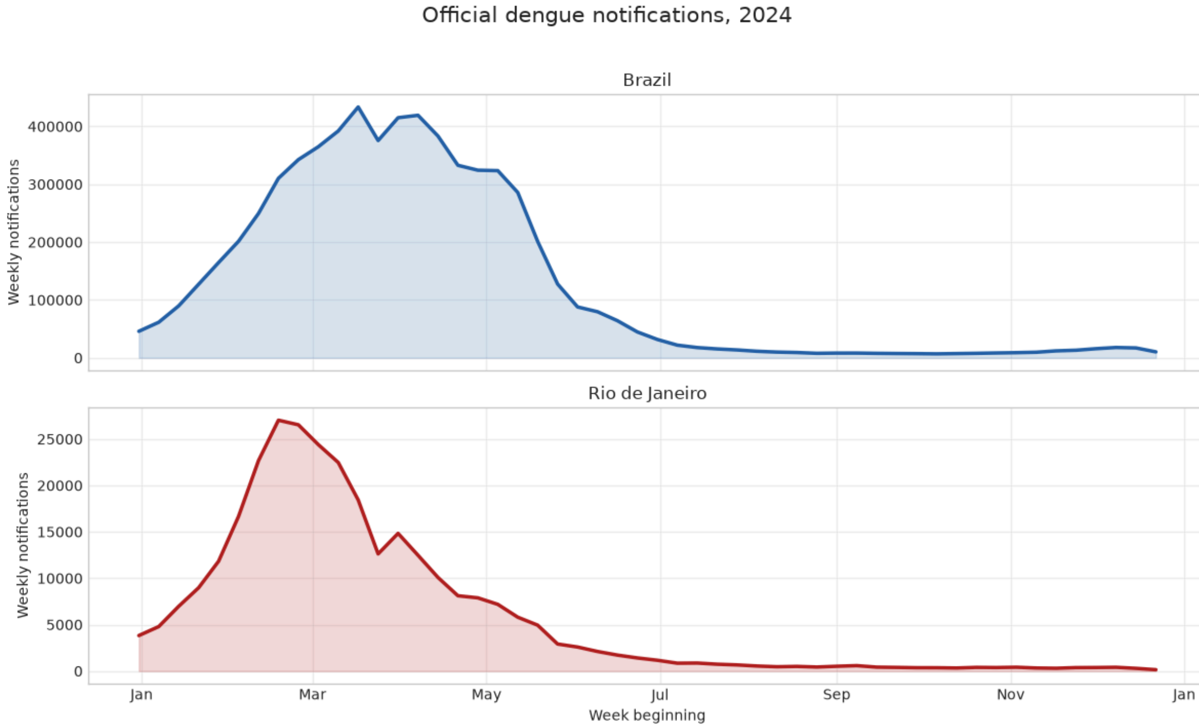
In [4]:
```
data_2024 = analysis.query('year == 2024').copy()
data_2024['peak_index'] = data_2024.groupby('geography')['sinan_notified_cas

fig, ax = plt.subplots(figsize=(11, 5))
for geography, group in data_2024.groupby('geography'):
    ax.plot(group['date'], group['peak_index'], lw=2.8, color=COLORS[geograp
ax.set(title="Dengue notifications in 2024: timing relative to each geograph
       xlabel='Week beginning', ylabel='Weekly notifications (peak = 100)')
ax.xaxis.set_major_locator(mdates.MonthLocator(interval=2))
ax.xaxis.set_major_formatter(mdates.DateFormatter('%b'))
ax.legend(frameon=False)
fig.tight_layout()
plt.show()
```



Step 4 — Show the absolute weekly counts

After establishing timing, show absolute counts in separate panels. Separate y-axes avoid making Rio appear flat beside the national total while preserving the actual number of notifications in each geography.

In [5]:
```
fig, axes = plt.subplots(2, 1, figsize=(12, 7), sharex=True)
for ax, geography in zip(axes, ['Brazil', 'Rio de Janeiro']):
    group = data_2024.query('geography == @geography')
    ax.fill_between(group['date'], group['sinan_notified_cases'], color=COLO
    ax.plot(group['date'], group['sinan_notified_cases'], color=COLORS[geogr
    ax.set(title=geography, ylabel='Weekly notifications')
    ax.grid(alpha=0.3)
axes[-1].set_xlabel('Week beginning')
axes[-1].xaxis.set_major_locator(mdates.MonthLocator(interval=2))
axes[-1].xaxis.set_major_formatter(mdates.DateFormatter('%b'))
fig.suptitle('Official dengue notifications, 2024', y=1.02, fontsize=15)
fig.tight_layout()
plt.show()
```

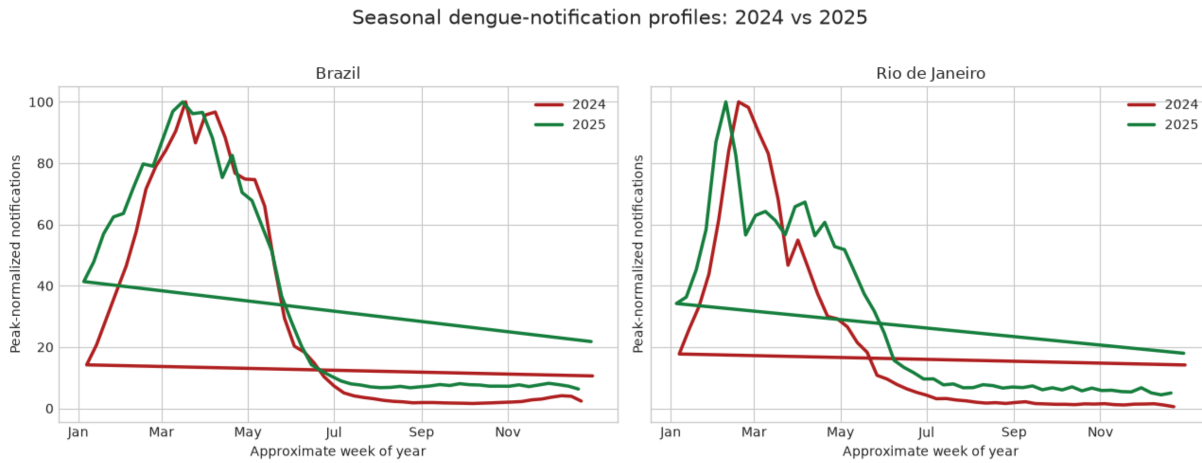


Step 5 — Compare seasons within each geography

This view answers a different question: how did the 2025 seasonal profile compare with 2024 within the same geography? Calendar position is used on the x-axis, and each year is indexed to its own peak. Do not interpret this as an incidence rate; denominators are not included in these files.

In [6]:

```
seasonal = analysis.copy()
seasonal['month_day'] = seasonal['date'].dt.strftime('%m-%d')
seasonal['peak_index'] = seasonal.groupby(['geography', 'year'])['sinan_noti

fig, axes = plt.subplots(1, 2, figsize=(13, 4.8), sharey=True)
for ax, geography in zip(axes, ['Brazil', 'Rio de Janeiro']):
    for year, group in seasonal.query('geography == @geography').groupby('ye
        ax.plot(group['date'].dt.dayofyear, group['peak_index'], lw=2.5, col
    ax.set(title=geography, xlabel='Approximate week of year', ylabel='Peak-
    ax.set_xticks([1, 60, 121, 182, 244, 305])
    ax.set_xticklabels(['Jan', 'Mar', 'May', 'Jul', 'Sep', 'Nov'])
    ax.legend(frameon=False)
fig.suptitle('Seasonal dengue-notification profiles: 2024 vs 2025', y=1.02,
fig.tight_layout()
plt.show()
```



Step 6 — Summarize the descriptive evidence

A compact table gives you numbers to cite while presenting. Totals exclude the boundary weeks identified above. The peak week is the week beginning date with the largest number of notifications.

In [7]:

```
summary = (analysis.groupby(['geography', 'year'])
           .agg(total_notifications=('sinan_notified_cases', 'sum'),
                peak_week=('date', lambda x: x.loc[x.index[analysis.loc[x.ir
                peak_notifications=('sinan_notified_cases', 'max'))
           .reset_index())
summary['peak_week'] = summary['peak_week'].dt.strftime('%Y-%m-%d')
summary
```

Out[7]:

	geography	year	total_notifications	peak_week	peak_notifications
0	Brazil	2024	6559686	2024-03-17	433513
1	Brazil	2025	1629233	2025-03-16	90265
2	Rio de Janeiro	2024	301812	2024-02-18	27026
3	Rio de Janeiro	2025	29351	2025-02-09	1997

Suggested closing script

- The figures describe notifications by **symptom-onset week**, not a causal relationship or a final confirmed-case count.
- Brazil and Rio show a pronounced early-year dengue wave in the archived data; peak-normalized plots are useful for comparing timing across unequal population sizes.
- Use the absolute-count panels for burden and the indexed panels for timing/shape—do not mix the two interpretations.
- The endpoint weeks are handled conservatively because calendar-year extracts do not contain a complete cross-year week.

- A next analytic step would require population denominators for rates, case-status fields for confirmed cases, or a pre-specified nowcasting design. Those additions are intentionally outside this DATASUS-only descriptive workflow.

Provenance. Brazil: `wikipedia/brazil_sinan_dengue_2024_weekly.csv` and `...2025...`; Rio: `rj_trends/rio_dengue_trends_vs_sinan_2024_weekly.csv` and `rio_dengue_sinan_2025_weekly.csv`. These archived series were derived from Brazilian Ministry of Health Open Data SUS / SINAN dengue line lists.

Part II — Online attention signals and nowcasts

The next figures answer a separate question from the descriptive DATASUS analysis: can contemporaneous online attention approximate the same week's SINAN notifications?

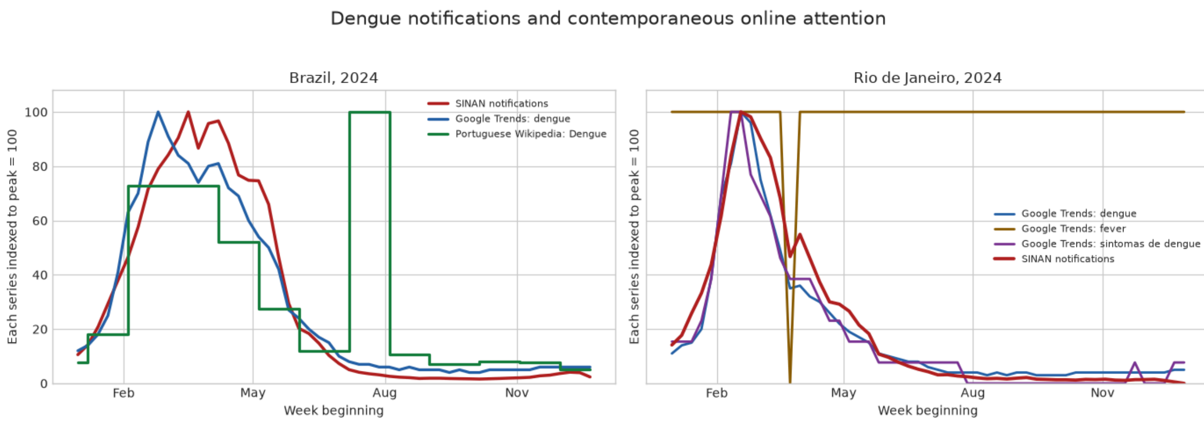
This is a **nowcast**, not an advance forecast. Google searches occur during the week of interest, and monthly Wikipedia totals are only complete after month-end. Predictors are therefore not treated as information available far in advance.

Step 7 — Inspect the attention signals before modeling

Compare timing rather than raw scale. Google Trends is a normalized 0–100 index; Portuguese Wikipedia is monthly pageviews. SINAN notifications are also indexed to their own maxima. The Brazil panel uses the national search term and Portuguese Wikipedia; the Rio panel uses Rio search terms and the same Brazil-oriented Portuguese Wikipedia proxy.

In [8]:
```
from IPython.display import Image, display

wiki_monthly = pd.read_csv(BRAZIL_DIR / 'brazil_wikipedia_pt_2023_2025_month
brazil_trends_2024 = pd.read_csv(BRAZIL_DIR / 'brazil_google_trends_dengue_2
brazil_cases_2024 = pd.read_csv(BRAZIL_DIR / 'brazil_sinan_dengue_2024_weekl
rio_signal_2024 = pd.read_csv(RIO_DIR / 'rio_dengue_trends_vs_sinan_2024_wee

brazil_signals = brazil_cases_2024.merge(brazil_trends_2024, on='date')
brazil_signals = pd.merge_asof(brazil_signals.sort_values('date'), wiki_mont
brazil_signals = brazil_signals.query("date < '2024-12-29'").copy()

def peak_index(values):
    return 100 * values / values.max()

fig, axes = plt.subplots(1, 2, figsize=(14, 4.6), sharey=True)
axes[0].plot(brazil_signals['date'], peak_index(brazil_signals['sinan_notifi
axes[0].plot(brazil_signals['date'], peak_index(brazil_signals['dengue']), c
axes[0].step(brazil_signals['date'], peak_index(brazil_signals['dengue_pt'])
axes[0].set_title('Brazil, 2024')
axes[0].legend(frameon=False, fontsize=8)
```

```
for column, label, color in [('dengue', 'Google Trends: dengue', '#2563a8'),
    axes[1].plot(rio_signal_2024['date'], peak_index(rio_signal_2024[column]
axes[1].plot(rio_signal_2024['date'], peak_index(rio_signal_2024['sinan_noti
axes[1].set_title('Rio de Janeiro, 2024')
axes[1].legend(frameon=False, fontsize=8)

for ax in axes:
    ax.set(xlabel='Week beginning', ylabel='Each series indexed to peak = 10
    ax.xaxis.set_major_locator(mdates.MonthLocator(interval=3))
    ax.xaxis.set_major_formatter(mdates.DateFormatter('%b'))
fig.suptitle('Dengue notifications and contemporaneous online attention', y=
fig.tight_layout()
plt.show()
```



Step 8 — Specify the retrospective model comparison

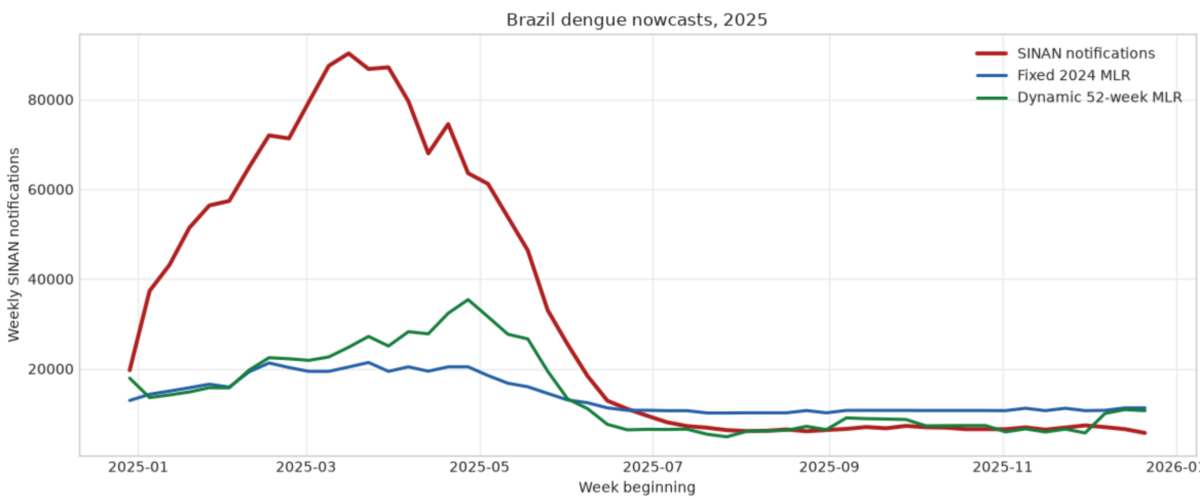
Geography	Fixed model	Dynamic model	Predictors
Brazil	Fit on 2024 only, then held fixed in 2025	Refit weekly on the previous 52 observed weeks	Same-week Brazil Google Trends dengue + monthly Portuguese Wikipedia Dengue pageviews
Rio	Fit on 2024 only, then held fixed in 2025	Refit weekly on the previous 52 observed weeks	Same-week Rio Google Trends (dengue, fever, sintomas de dengue) + the same monthly Portuguese Wikipedia proxy

The dynamic model can use notifications from earlier 2025 weeks, but never the current week's outcome. Both analyses exclude the partial week beginning 2025-12-28 from scoring.

Step 9 — Evaluate the Brazil nowcasts in 2025

The fixed model is the strict year-ahead test: its coefficients come only from 2024. The dynamic model is operationally different: it adapts to recent observed notifications. Compare trajectories first, then use MAE, RMSE, and correlation as descriptive accuracy summaries.

In [9]:
```
brazil_pred = pd.read_csv(BRAZIL_DIR / 'brazil_dengue_2025_mlr_predictions.c
brazil_metrics = pd.read_csv(BRAZIL_DIR / 'brazil_dengue_2025_mlr_metrics.cs

fig, ax = plt.subplots(figsize=(12, 5))
ax.plot(brazil_pred['date'], brazil_pred['sinan_notified_cases'], color='#b2
ax.plot(brazil_pred['date'], brazil_pred['fixed_2024_mlr_prediction'], color
ax.plot(brazil_pred['date'], brazil_pred['dynamic_52_week_mlr_prediction'],
ax.set(title='Brazil dengue nowcasts, 2025', xlabel='Week beginning', ylabel
ax.legend(frameon=False)
ax.grid(alpha=0.3)
fig.tight_layout()
plt.show()

display(brazil_metrics.round({'mae': 0, 'rmse': 0, 'pearson_r': 2, 'predicte
```



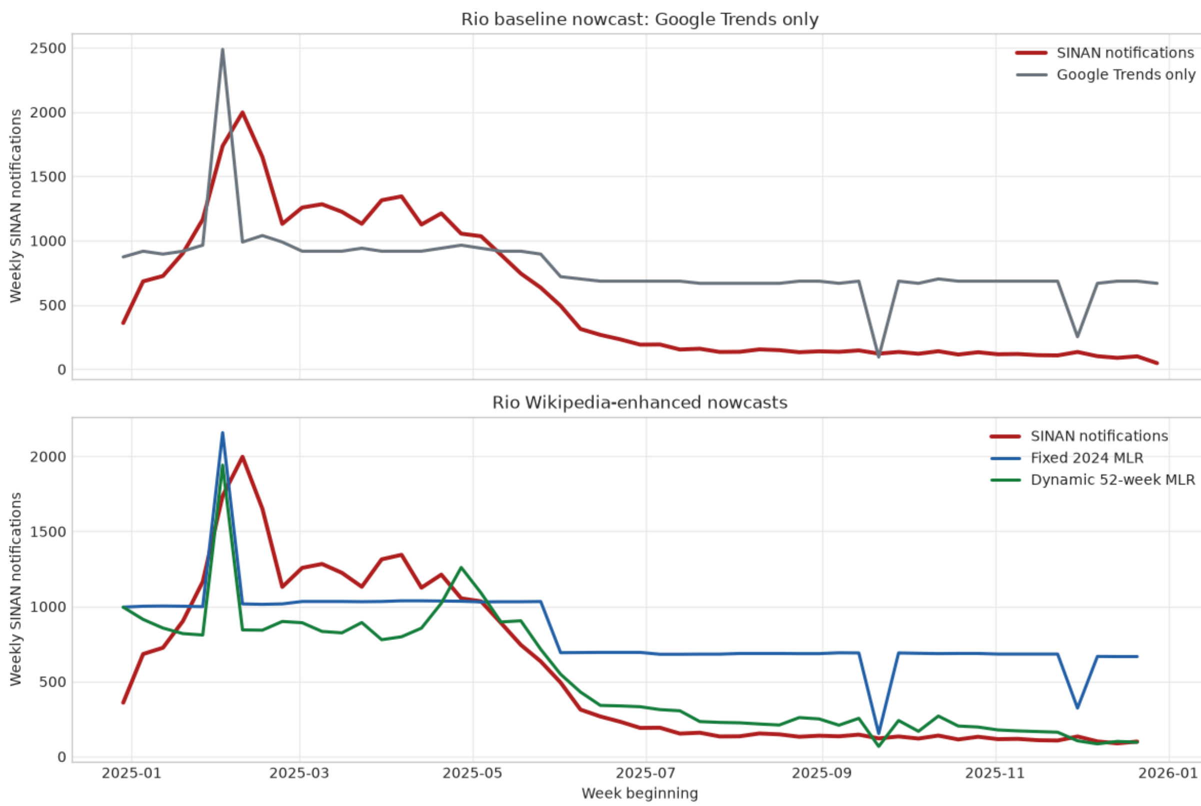
	model	mae	rmse	pearson_r	predicted_total
0	Fixed 2024 MLR	21294.0	31286.0	0.98	721956.0
1	Dynamic 52-week MLR	18131.0	28272.0	0.89	733816.0

Step 10 — Evaluate the Rio nowcasts in 2025

For Rio, retain the original Google-Trends-only model as a baseline, then compare it with Wikipedia-enhanced fixed and dynamic models. Keeping that baseline prevents us from attributing an apparent improvement to Wikipedia without a matched comparison.

In [10]:
```
rio_google_only = pd.read_csv(RIO_DIR / 'rio_dengue_2025_nowcast_predictions
rio_pred = pd.read_csv(RIO_DIR / 'rio_dengue_2025_mlr_dynamic_predictions.cs
rio_metrics = pd.read_csv(RIO_DIR / 'rio_dengue_2025_mlr_dynamic_metrics.csv

fig, axes = plt.subplots(2, 1, figsize=(12, 8), sharex=True)
axes[0].plot(rio_google_only['date'], rio_google_only['sinan_notified_cases'
axes[0].plot(rio_google_only['date'], rio_google_only['predicted_notified_ca
axes[0].set(title='Rio baseline nowcast: Google Trends only', ylabel='Weekly
axes[0].legend(frameon=False)
```

```
axes[1].plot(rio_pred['date'], rio_pred['sinan_notified_cases'], color='#b22
axes[1].plot(rio_pred['date'], rio_pred['fixed_mlr_prediction'], color='#256
axes[1].plot(rio_pred['date'], rio_pred['dynamic_52_week_prediction'], color
axes[1].set(title='Rio Wikipedia-enhanced nowcasts', xlabel='Week beginning'
axes[1].legend(frameon=False)
for ax in axes:
    ax.grid(alpha=0.3)
fig.tight_layout()
plt.show()

display(rio_metrics.round({'mae': 0, 'rmse': 0, 'pearson_r': 2, 'predicted_t
```



	model	mae	rmse	pearson_r	predicted_total
0	Fixed 2024 MLR	404.0	454.0	0.78	43295.0
1	Dynamic 52-week MLR	186.0	285.0	0.86	27586.0

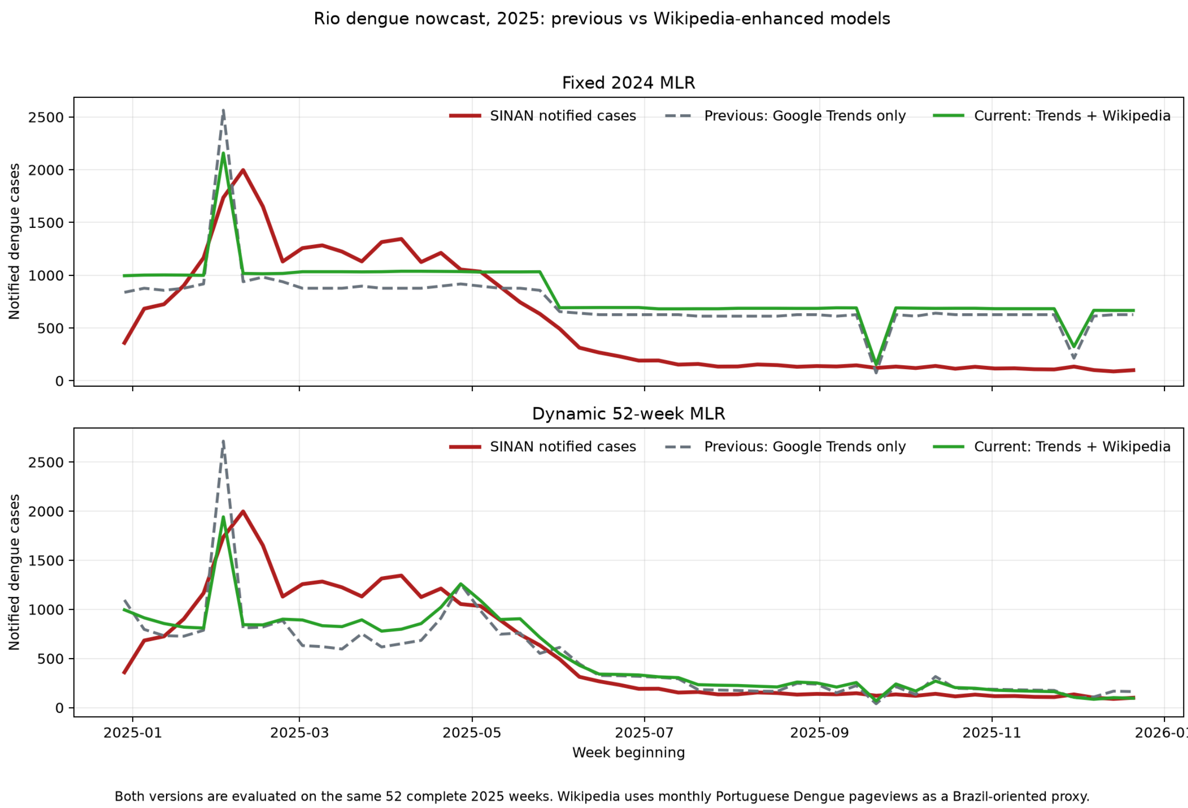
Step 11 — Isolate the value of Wikipedia in Rio

This is the matched evaluation: Google-Trends-only and Trends-plus-Wikipedia versions are scored on the same 52 complete 2025 weeks. Lower MAE/RMSE and higher correlation indicate a closer retrospective match to the SINAN notifications. The comparison is associational; it does not establish that pageviews cause changes in notifications.

In [11]:
```
rio_version_metrics = pd.read_csv(RIO_DIR / 'rio_dengue_2025_nowcast_versior
display(rio_version_metrics.round({'mae': 0, 'rmse': 0, 'pearson_r': 2}))
```

```
display(Image(filename=str(RIO_DIR / 'rio_dengue_2025_nowcast_version_compar
```

	model	version	mae	rmse	pearson_r
0	Fixed 2024 MLR	Google Trends only	392.0	438.0	0.68
1	Fixed 2024 MLR	Trends + Wikipedia	404.0	454.0	0.78
2	Dynamic 52-week MLR	Google Trends only	226.0	360.0	0.75
3	Dynamic 52-week MLR	Trends + Wikipedia	186.0	285.0	0.86



Final presentation takeaways

1. **Start with DATASUS/SINAN.** The descriptive plots establish seasonal notification patterns before any model.
2. **Separate timing from burden.** Peak-normalized figures compare shapes; absolute-count panels convey notification volume.
3. **Name the prediction target precisely.** Models estimate same-week notified records by symptom-onset week, not final confirmed or future cases.
4. **Keep comparisons fair.** Fixed 2024 models test temporal transfer; rolling 52-week models adapt using prior observed weeks.
5. **Treat Wikipedia as a Brazil-oriented attention proxy.** It is monthly, not Rio-specific, and its apparent Rio value is assessed only through the matched 2025 comparison.

Sources. Outcomes are the archived DATASUS/SINAN weekly series. Predictors and model outputs are the saved Google Trends and Portuguese Wikipedia extracts in rj_trends and wikipedia. The final week crossing into 2026 is excluded from model scores.

Part III — Does notification-system activity track disease-specific public attention?

Hypothesis

The first formal notification of an arboviral illness in SINAN may be correlated with disease-specific information seeking. To examine that hypothesis, we compare weekly Brazil-wide notification dates for **dengue**, **chikungunya**, and **Zika** with Portuguese Google searches and Portuguese Wikipedia pageviews for each disease.

This is an observational salience analysis. It asks whether signals move together; it does not identify the cause of notifications or prove that online attention affects diagnosis.

Step 12 — Align the outcome with the hypothesis

This section changes the outcome definition used in Parts I and II. The outcome is now **DT_NOTIFIC**, the date on which a record is notified in SINAN, aggregated to Sunday–Saturday weeks. That is the appropriate timestamp for a question about initial information-system registration.

Disease	Official SINAN source	Portuguese Google term	Portuguese Wikipedia article
Dengue	Open Data SUS dengue line list	dengue	Dengue
Chikungunya	Open Data SUS chikungunya line list	chikungunya	Chicungunha
Zika	Open Data SUS Zika-virus line list	zika	Febre zika

The 2024 and 2025 annual extracts include boundary weeks. For fair evaluation, the model trains on 2024 surveillance weeks beginning 2023-12-31 through 2024-12-22 and evaluates the 52 complete 2025 weeks beginning 2024-12-29 through 2025-12-21.

In [12]:
```
import numpy as np

PART3_DIR = ROOT / 'part3_arboviruses'
arbovirus = pd.read_csv(PART3_DIR / 'brazil_arbovirus_notification_attentior
arbovirus['analysis_year'] = np.where(arbovirus['date'].between('2023-12-31'
                                      np.where(arbovirus['date'].between('20
arbovirus_analysis = arbovirus.dropna(subset=['analysis_year']).copy()
arbovirus_analysis['analysis_year'] = arbovirus_analysis['analysis_year'].as

quality = (arbovirus_analysis.groupby(['disease', 'analysis_year'])
           .agg(weeks=('date', 'size'),
                first_week=('date', 'min'),
                last_week=('date', 'max'),
                notifications=('sinan_notifications', 'sum'),
                missing_signals=('google_trends', lambda x: x.isna().sum()))
           .reset_index())
quality
```

Out[12]:

	disease	analysis_year	weeks	first_week	last_week	notifications	mis
0	chikungunya	2024	52	2023-12-31	2024-12-22	425040	
1	chikungunya	2025	52	2024-12-29	2025-12-21	250856	
2	dengue	2024	52	2023-12-31	2024-12-22	6563899	
3	dengue	2025	52	2024-12-29	2025-12-21	1631977	
4	zika	2024	52	2023-12-31	2024-12-22	43155	
5	zika	2025	52	2024-12-29	2025-12-21	25723	

Step 13 — Inspect disease-specific notification and attention patterns

All three lines in each panel are scaled to that disease's own peak. This allows a within-disease timing comparison across different units: SINAN notifications, Google Trends interest, and monthly Wikipedia pageviews. It does **not** compare absolute disease burden across diseases.

In [13]:
```
def index_to_peak(values):
    return 100 * values / values.max()

signal_labels = {
    'sinan_notifications': ('SINAN notification date', '#b22222'),
    'google_trends': ('Google Trends, Portuguese term', '#2563a8'),
    'wikipedia_pageviews': ('Portuguese Wikipedia pageviews', '#16803c'),
}
fig, axes = plt.subplots(3, 1, figsize=(12, 10), sharex=True)
```

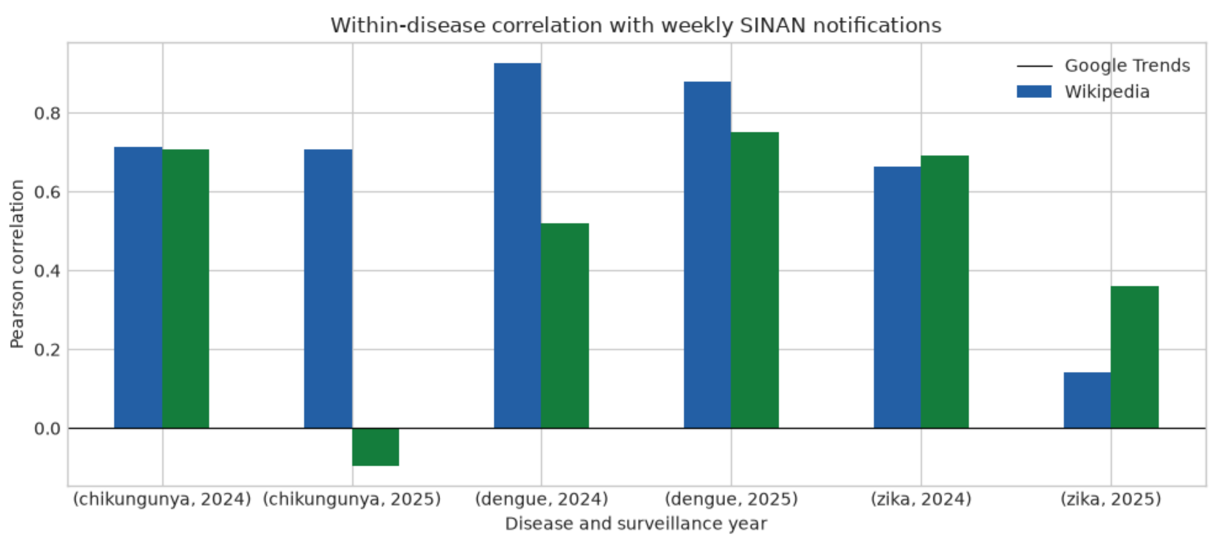
```
for ax, disease in zip(axes, ['dengue', 'chikungunya', 'zika']):
    group = arbovirus_analysis.query("analysis_year == 2024 and disease == @
    for variable, (label, color) in signal_labels.items():
        draw = ax.step if variable == 'wikipedia_pageviews' else ax.plot
        if variable == 'wikipedia_pageviews':
            draw(group['date'], index_to_peak(group[variable]), where='post'
        else:
            draw(group['date'], index_to_peak(group[variable]), color=color,
    ax.set(title=disease.capitalize(), ylabel='Each series: peak = 100', yli
    ax.legend(frameon=False, ncol=3, fontsize=8)
    ax.grid(alpha=0.25)
axes[-1].set_xlabel('Notification week beginning')
axes[-1].xaxis.set_major_locator(mdates.MonthLocator(interval=2))
axes[-1].xaxis.set_major_formatter(mdates.DateFormatter('%b'))
fig.suptitle('Notification-system activity and disease-specific public atter
fig.tight_layout()
plt.show()
```

Step 14 — Quantify the association, rather than relying only on visual alignment

Pearson correlation captures linear co-movement; Spearman correlation captures agreement in weekly rank/order. Each correlation is within disease and surveillance year. These values are descriptive because outbreaks, reporting practices, media coverage, and information seeking can all be common causes.

```
In [14]: correlations = pd.read_csv(PARTS_DIR / 'brazil_arbovirus_attention_correlati
display(correlations.round({'pearson_r': 2, 'spearman_r': 2}))

plot_data = correlations.pivot(index=['disease', 'year'], columns='signal',
ax = plot_data.plot.bar(figsize=(10, 4.5), color=['#2563a8', '#16803c'])
ax.axhline(0, color='black', lw=0.8)
ax.set(title='Within-disease correlation with weekly SINAN notifications',
       xlabel='Disease and surveillance year', ylabel='Pearson correlation')
ax.legend(['Google Trends', 'Wikipedia'], frameon=False)
ax.tick_params(axis='x', rotation=0)
ax.figure.tight_layout()
plt.show()
```

	year	disease	signal	pearson_r	spearman_r
0	2024	chikungunya	google_trends	0.71	0.69
1	2024	chikungunya	wikipedia_pageviews	0.71	0.70
2	2024	dengue	google_trends	0.93	0.88
3	2024	dengue	wikipedia_pageviews	0.52	0.63
4	2024	zika	google_trends	0.66	0.63
5	2024	zika	wikipedia_pageviews	0.69	0.65
6	2025	chikungunya	google_trends	0.71	0.74
7	2025	chikungunya	wikipedia_pageviews	-0.09	0.23
8	2025	dengue	google_trends	0.88	0.79
9	2025	dengue	wikipedia_pageviews	0.75	0.64
10	2025	zika	google_trends	0.14	0.16
11	2025	zika	wikipedia_pageviews	0.36	0.37



Step 15 — Compare fixed and dynamic seasonal nowcasts

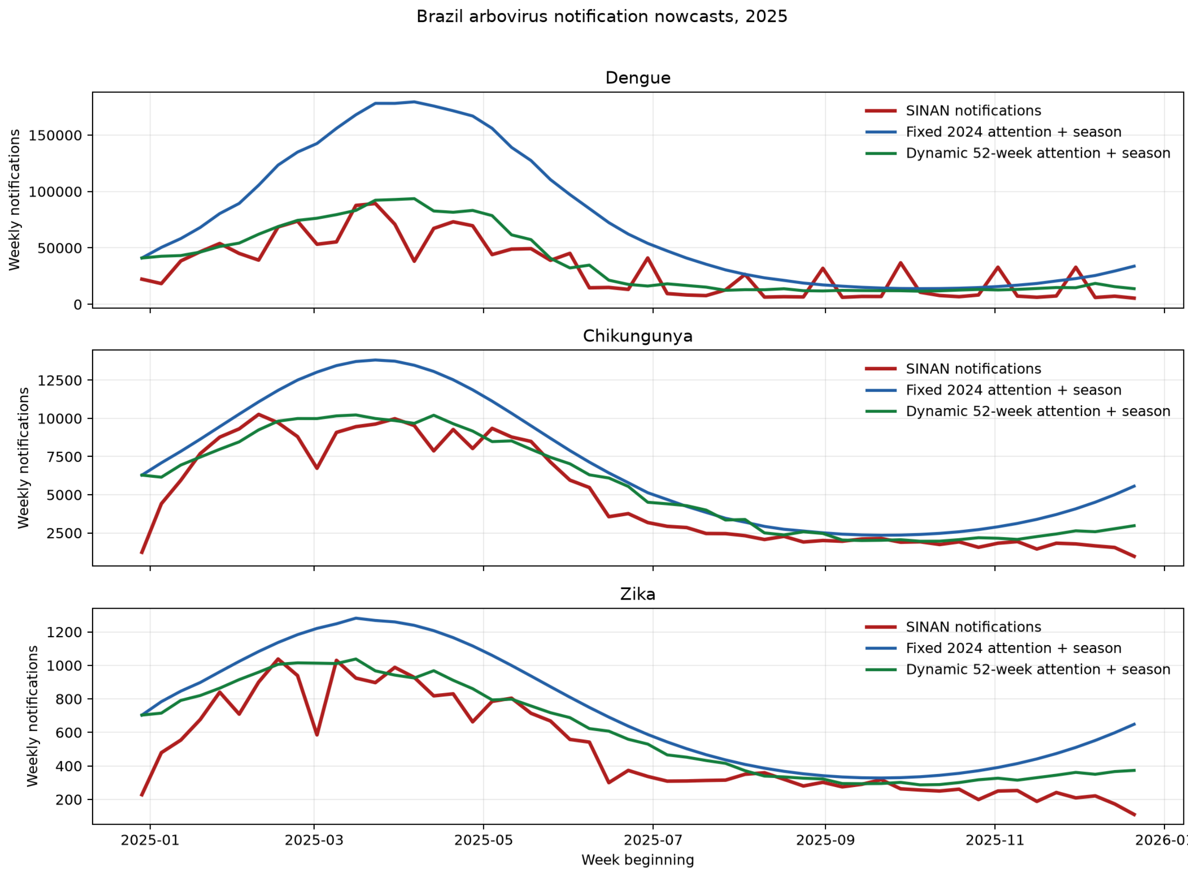
Both models are pooled ridge regressions predicting log weekly notifications from the disease-specific Portuguese Google Trends value, Portuguese Wikipedia pageviews, disease indicators, and cyclic week-of-year terms. Disease-specific sine/cosine interactions allow each disease to have a different seasonal curve. Ridge regularization stabilizes correlated attention predictors.

- **Fixed 2024 attention + season:** fit using 2024 surveillance weeks only, then held fixed for 2025. This is the temporal-transfer test.
- **Dynamic 52-week attention + season:** before each 2025 prediction, refit using only the preceding 52 complete notification weeks across all diseases. It can use earlier 2025 notifications, but never the current week's outcome.

The dynamic result is a retrospective nowcast comparison, not an advance forecast or causal effect estimate.

In [15]:
```
arbovirus_metrics = pd.read_csv(PART3_DIR / 'brazil_arbovirus_2025_attentior
display(arbovirus_metrics.round({'mae': 0, 'rmse': 0, 'pearson_r': 2, 'obser
display(Image(filename=str(PART3_DIR / 'brazil_arbovirus_2025_attention_prec
```

	model	disease	mae	rmse	pearson_r	observed_total	predicted_
0	Fixed 2024 attention + season	chikungunya	2059.0	2541.0	0.94	250856	357
1	Fixed 2024 attention + season	dengue	42874.0	56169.0	0.86	1631977	3732
2	Fixed 2024 attention + season	zika	227.0	265.0	0.91	25723	37
3	Fixed 2024 attention + season	pooled	15053.0	32463.0	0.92	1908556	4127
4	Dynamic 52-week attention + season	chikungunya	913.0	1281.0	0.95	250856	288
5	Dynamic 52-week attention + season	dengue	11585.0	15390.0	0.87	1631977	1950
6	Dynamic 52-week attention + season	zika	109.0	149.0	0.92	25723	31
7	Dynamic 52-week attention + season	pooled	4202.0	8917.0	0.93	1908556	2269



Outcome: weekly SINAN notification date. Predictors: disease-specific Portuguese attention signals plus cyclic week-of-year seasonality.

Part III interpretation and suggested presentation script

- **Use the word association.** The analysis finds that disease-specific online attention and weekly notification-system activity often co-vary, especially for dengue and chikungunya in these archived periods.
- **Zika is a useful contrast.** Its weaker 2025 correlations/model fit show that a named online signal is not automatically a reliable proxy for every low- or differently-salient disease.
- **Notification timing is not disease onset.** DT_NOTIFIC can reflect care seeking, clinical suspicion, laboratory/testing workflows, and reporting delays in addition to transmission.
- **Salience is disease-specific but not perfectly diagnostic-specific.** Search terms and articles can be used by the public, clinicians, media, and students; they do not identify who searched or why.
- **The model is a retrospective 2024-to-2025 check, not a causal or operational deployment claim.** A next step would pre-register lag structures, reporting-delay handling, and a prospective evaluation.

Data provenance. The outcome tables were downloaded from the official Brazilian Ministry of Health Open Data SUS / SINAN annual line lists for 2024 and

2025: dengue (CSV), chikungunya (CSV), and Zika virus (JSON). Portuguese Google Trends terms were retrieved together for Brazil over 2024–2025; Portuguese Wikipedia monthly pageviews use the disease-specific articles listed above.

Part IV — Mechanistic comparator on the same diseases and periods

The attention models are predictive models. To make the comparison more informative, this section adds a disease-specific **seasonal renewal model**: a simple transmission-inspired comparator with no Google Trends or Wikipedia inputs.

It is trained on the same 2024 notification weeks and evaluated on the identical 52 complete 2025 weeks for dengue, chikungunya, and Zika.

Step 16 — Define the mechanistic seasonal-renewal model

For each disease, expected weekly activity is generated from recent activity, seasonal transmission, and depletion of an estimated effective pool:

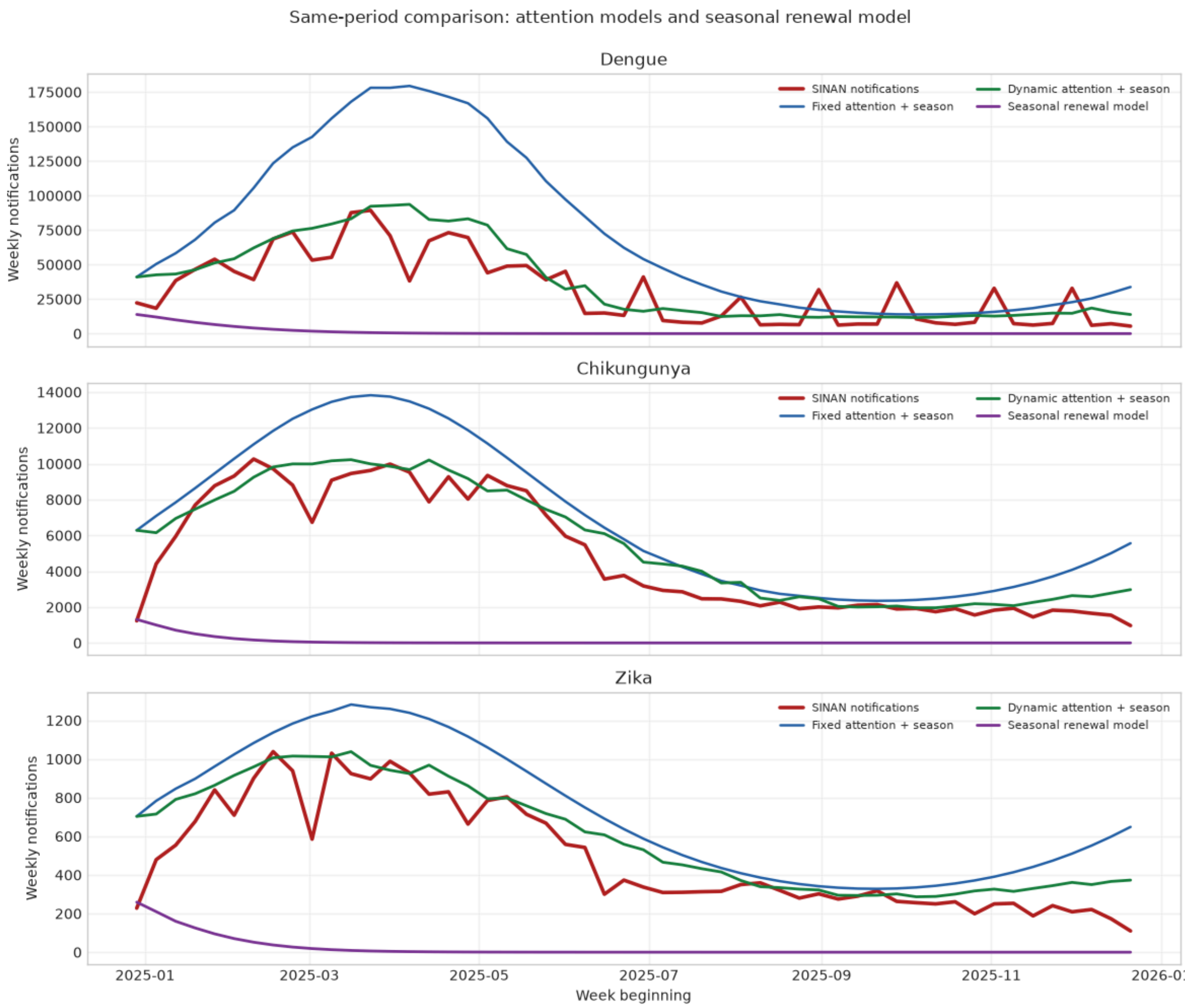
expected notifications at week t = seasonal reproduction factor at t × weighted recent notifications × remaining effective-pool fraction.

- The recent-notification weights are 0.55, 0.30, and 0.15 for the preceding one, two, and three weeks.
- The reproduction factor uses sine/cosine week-of-year terms, learned from 2024 only.
- The effective pool is fitted as a notification-scale parameter; it is **not** a population size or a direct susceptible-person estimate.
- 2025 values are generated recursively. Once forecasting starts, the model never sees an observed 2025 notification count.

This is mechanistic in its renewal/process structure, but it remains a proxy model because the target is notification date rather than infection onset.

In [16]:
```
mechanistic_pred = pd.read_csv(PART3_DIR / 'brazil_arbovirus_2025_mechanisti
attention_pred = pd.read_csv(PART3_DIR / 'brazil_arbovirus_2025_attention_pr
comparison_pred = attention_pred.merge(
    mechanistic_pred[['date', 'disease', 'seasonal_renewal_prediction']],
    on=['date', 'disease'], validate='one_to_one'
)

fig, axes = plt.subplots(3, 1, figsize=(12, 10), sharex=True)
```

```
for ax, disease in zip(axes, ['dengue', 'chikungunya', 'zika']):
    group = comparison_pred.query('disease == @disease')
    ax.plot(group['date'], group['sinan_notifications'], color='#b22222', lw
    ax.plot(group['date'], group['fixed_2024_attention_prediction'], color='
    ax.plot(group['date'], group['dynamic_52_week_seasonal_prediction'], col
    ax.plot(group['date'], group['seasonal_renewal_prediction'], color='#7b3
    ax.set(title=disease.capitalize(), ylabel='Weekly notifications')
    ax.grid(alpha=0.25)
    ax.legend(frameon=False, ncol=2, fontsize=8)
axes[-1].set_xlabel('Week beginning')
fig.suptitle('Same-period comparison: attention models and seasonal renewal
fig.tight_layout()
plt.show()
```



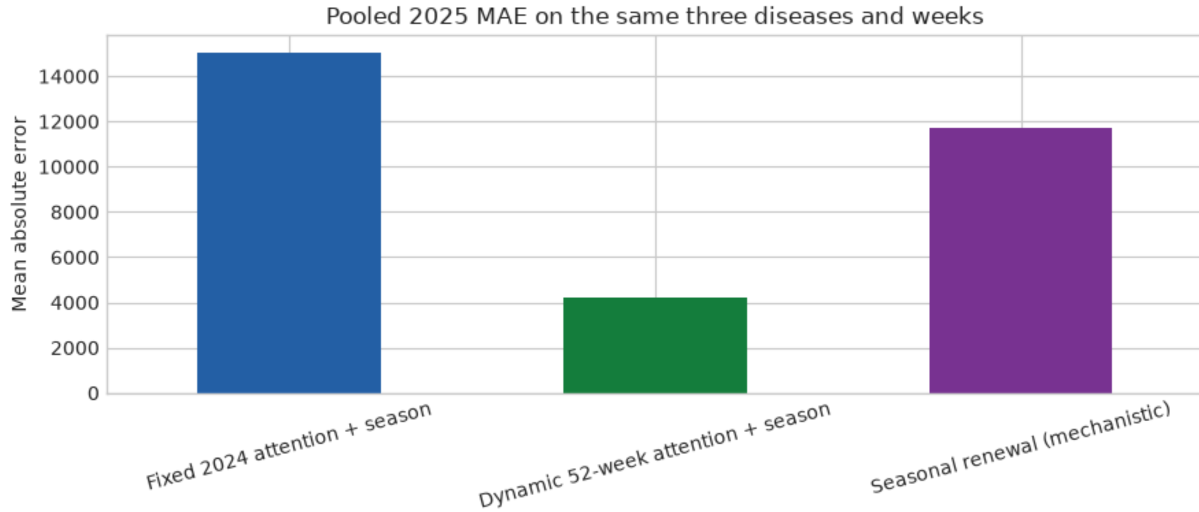
Step 17 — Compare forecast accuracy on the same 2025 weeks

All rows below use the same notification-week outcome and hold out period. The fixed attention model and renewal model use 2024 data only. The dynamic attention model has a different operational advantage: it refits with earlier 2025 notifications, but never the current week's outcome.

In [17]:

```
attention_metrics = pd.read_csv(PART3_DIR / 'brazil_arbovirus_2025_attention
mechanistic_metrics = pd.read_csv(PART3_DIR / 'brazil_arbovirus_2025_mechani
mechanistic_metrics.insert(0, 'model', 'Seasonal renewal (mechanistic)')
model_comparison = pd.concat([attention_metrics, mechanistic_metrics], ignor
display(model_comparison.round({'mae': 0, 'rmse': 0, 'pearson_r': 2, 'observ

pooled = model_comparison.query("disease == 'pooled'").copy()
ax = pooled.plot.bar(x='model', y='mae', color=['#2563a8', '#16803c', '#7b32
ax.set(title='Pooled 2025 MAE on the same three diseases and weeks', xlabel=
ax.tick_params(axis='x', rotation=15)
ax.figure.tight_layout()
plt.show()
```

	model	disease	mae	rmse	pearson_r	observed_total	pred
0	Fixed 2024 attention + season	chikungunya	2059.0	2541.0	0.94	250856	
1	Fixed 2024 attention + season	dengue	42874.0	56169.0	0.86	1631977	
2	Fixed 2024 attention + season	zika	227.0	265.0	0.91	25723	
3	Fixed 2024 attention + season	pooled	15053.0	32463.0	0.92	1908556	
4	Dynamic 52-week attention + season	chikungunya	913.0	1281.0	0.95	250856	
5	Dynamic 52-week attention + season	dengue	11585.0	15390.0	0.87	1631977	
6	Dynamic 52-week attention + season	zika	109.0	149.0	0.92	25723	
7	Dynamic 52-week attention + season	pooled	4202.0	8917.0	0.93	1908556	
8	Seasonal renewal (mechanistic)	chikungunya	4739.0	5746.0	0.04	250856	
9	Seasonal renewal (mechanistic)	dengue	30000.0	38669.0	0.15	1631977	
10	Seasonal renewal (mechanistic)	zika	475.0	549.0	0.10	25723	
11	Seasonal renewal (mechanistic)	pooled	11738.0	22573.0	0.32	1908556	



How to present this comparison

- **The comparison is fair on period and outcome.** Every model is evaluated against the same 52 complete 2025 notification weeks for the same three national disease series.
- **The mechanistic model is intentionally parsimonious.** It represents renewal, seasonality, and effective-pool depletion, but does not contain reporting delays, care seeking, diagnosis, media effects, vector abundance, or the online signals.
- **The attention models are not mechanistic.** Their stronger retrospective fit demonstrates predictive association with the notification system, not disease-transmission causality.
- **The gap is substantively informative.** A notification-date target combines transmission with health-system processes; a fully mechanistic infection model would need onset dates, reporting-delay distributions, population/immune structure, and ideally vector/climate data.
- **Next step.** A stronger mechanistic comparison would be a jointly fitted latent-infection observation model: seasonal transmission drives latent infections, while a disease-specific delay/ascertainment model maps infections to SINAN notifications.

Part V — Joint latent-infection observation model

This model separates two layers that the renewal comparator combined:

1. A disease-specific latent infection process follows seasonal renewal transmission with susceptible depletion.

2. Weekly SINAN notifications are a delayed, ascertained observation of those latent infections.

It is fitted separately for dengue, chikungunya, and Zika using 2024 notification weeks, then recursively forecast over the same complete 2025 weeks used in every previous comparison.

Step 18 — Specify the latent process and observation process

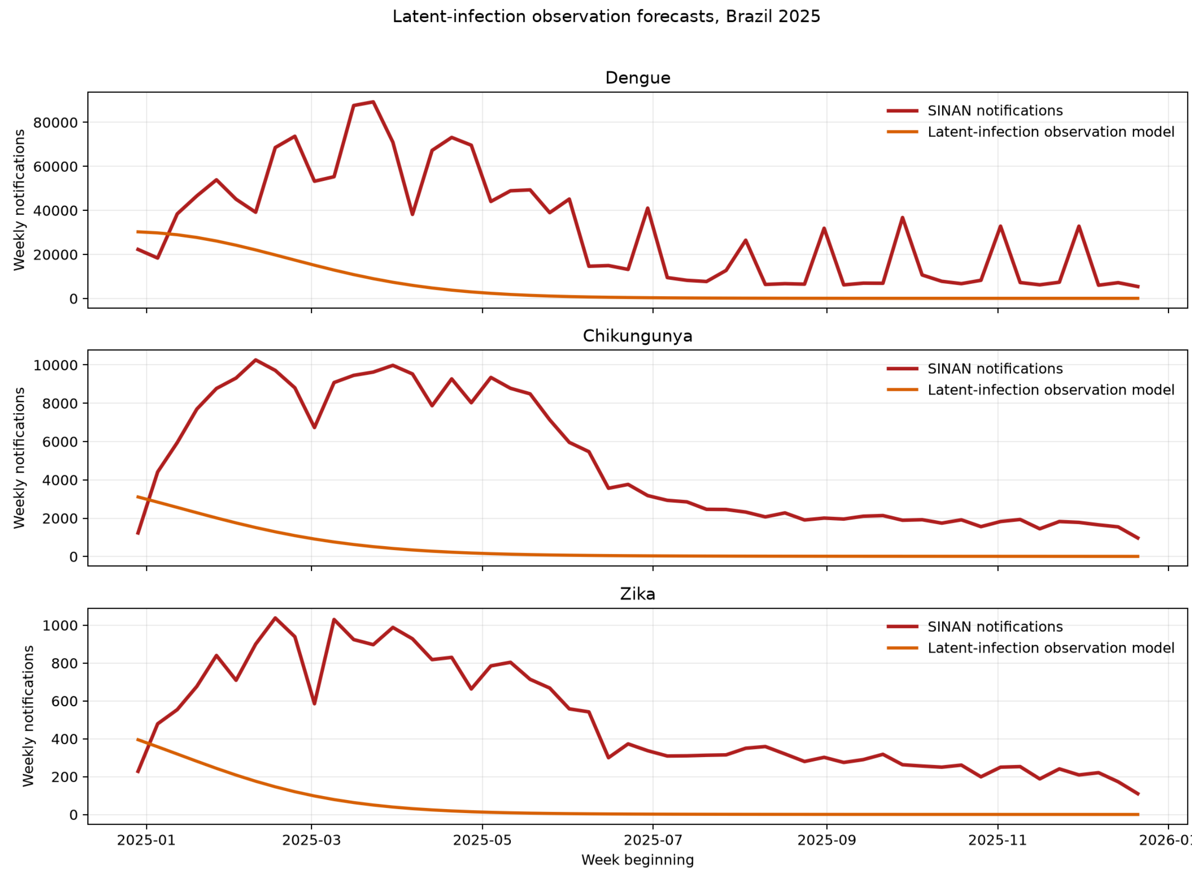
Latent infections. Each week's infections arise from a weighted three-week renewal pressure, a sine/cosine seasonal reproduction term, and depletion of the national susceptible pool. The susceptible-population anchor is the 2022 IBGE Brazil Census total of 203,062,512 residents.

Observation. Expected notifications equal a disease-specific reporting fraction times a disease-specific distribution of reporting delays from zero to three weeks.

Identification warning. Notification counts alone do not cleanly separate the number of latent infections from the reporting fraction. A weak regularization anchors the reporting fraction, so the fitted fractions are sensitivity parameters of this model—not estimates of real surveillance coverage. This makes the model useful as a structured comparator, not as a direct infection-burden estimator.

In [18]:
```
latent_pred = pd.read_csv(PART3_DIR / 'brazil_arbovirus_2025_latent_observat
latent_metrics = pd.read_csv(PART3_DIR / 'brazil_arbovirus_2025_latent_obser
latent_parameters = pd.read_csv(PART3_DIR / 'brazil_arbovirus_latent_observa

display(latent_parameters.round(3))
display(Image(filename=str(PART3_DIR / 'brazil_arbovirus_2025_latent_observa
```

	disease	ascertainment_fraction	same_week_delay_weight	one_week_dela
0	dengue	0.083	0.981	
1	chikungunya	0.006	0.678	
2	zika	0.001	1.000	



Fit on 2024 only. Latent infections: seasonal renewal; observations: disease-specific reporting fraction and 0–3 week delay distribution.

Step 19 — Compare all models on the same 2025 notification weeks

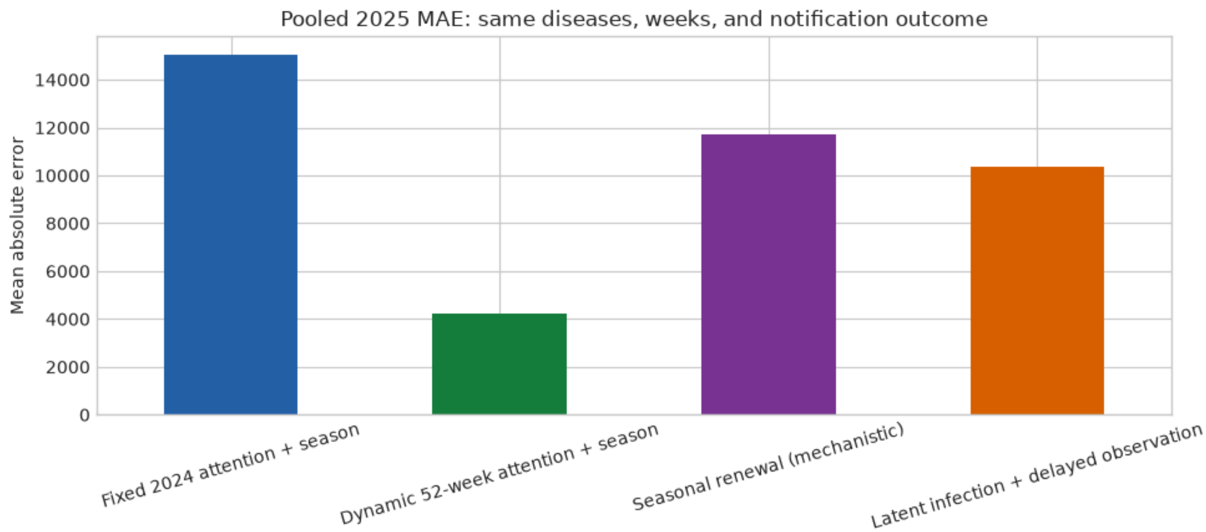
The latent-observation model and the seasonal renewal model are trained on 2024 only. The dynamic attention model is a rolling nowcast, so it has access to earlier observed 2025 notifications; it remains included because it represents the operational benchmark already considered in Part III.

In [19]:

```
all_model_metrics = pd.concat([
    attention_metrics,
    mechanistic_metrics.assign(model='Seasonal renewal (mechanistic)'),
    latent_metrics.assign(model='Latent infection + delayed observation'),
], ignore_index=True)
display(all_model_metrics.round({'mae': 0, 'rmse': 0, 'pearson_r': 2, 'obser

pooled_all = all_model_metrics.query("disease == 'pooled'").copy()
ax = pooled_all.plot.bar(x='model', y='mae',
                         color=['#2563a8', '#16803c', '#7b3294', '#d95f02'],
                         legend=False, figsize=(10, 4.5))
ax.set(title='Pooled 2025 MAE: same diseases, weeks, and notification outcom
       xlabel='', ylabel='Mean absolute error')
ax.tick_params(axis='x', rotation=16)
ax.figure.tight_layout()
plt.show()
```

	model	disease	mae	rmse	pearson_r	observed_total	pred
0	Fixed 2024 attention + season	chikungunya	2059.0	2541.0	0.94	250856	
1	Fixed 2024 attention + season	dengue	42874.0	56169.0	0.86	1631977	
2	Fixed 2024 attention + season	zika	227.0	265.0	0.91	25723	
3	Fixed 2024 attention + season	pooled	15053.0	32463.0	0.92	1908556	
4	Dynamic 52-week attention + season	chikungunya	913.0	1281.0	0.95	250856	
5	Dynamic 52-week attention + season	dengue	11585.0	15390.0	0.87	1631977	
6	Dynamic 52-week attention + season	zika	109.0	149.0	0.92	25723	
7	Dynamic 52-week attention + season	pooled	4202.0	8917.0	0.93	1908556	
8	Seasonal renewal (mechanistic)	chikungunya	4739.0	5746.0	0.04	250856	
9	Seasonal renewal (mechanistic)	dengue	30000.0	38669.0	0.15	1631977	
10	Seasonal renewal (mechanistic)	zika	475.0	549.0	0.10	25723	
11	Seasonal renewal (mechanistic)	pooled	11738.0	22573.0	0.32	1908556	
12	Latent infection + delayed observation	chikungunya	4441.0	5331.0	0.36	250856	
13	Latent infection + delayed observation	dengue	26159.0	33896.0	0.42	1631977	

	model	disease	mae	rmse	pearson_r	observed_total	pred
14	Latent infection + delayed observation	zika	449.0	515.0	0.31	25723	
15	Latent infection + delayed observation	pooled	10349.0	19813.0	0.57	1908556	



Interpretation of the latent-observation comparison

- The joint model improves on the simpler seasonal renewal comparator, showing that explicitly allowing a delayed, incomplete observation process matters for notification data.
- It still trails the attention models in this retrospective notification-nowcast task. That does **not** mean public-attention signals are causal; it means they track additional health-system and salience processes that the mechanistic model omits.
- The fitted zero-to-three-week delays and ascertainment fractions should not be reported as literal epidemiologic quantities. They are weakly identified from notifications alone and depend on the population anchor, delay support, renewal weights, and regularization.
- A stronger inferential implementation would add external information: onset dates, line-list reporting delays, serology or hospitalization data, vector/climate covariates, and prior distributions informed by disease-specific natural history.

Population anchor. The model uses the IBGE 2022 Census national total (203,062,512) only to set the susceptible-scale constraint; it does not imply every resident was immunologically susceptible to each disease.